Duality in Convex Optimisation

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1 Introduction

Informally, optimisation problems aim to maximise or minimise a specific quantity, subject to certain limitations. As a quick example, the packing of a suitcase, with the goal of maximising the total value of the packed items, and the limiting size of the suitcase. How should we decide which items to pack?

This real-world example demonstrates the key components of an optimisation problem:

- An objective.
- Constraints.
- Decision variables.

Definition 1.1 (Optimisation problem)**.** Formally, we define an *optimisation problem* as follows.
Let $x \in \mathbb{R}^n$, and $F, f_i, h_i : \mathbb{R}^n \to \mathbb{R}$.

$$\begin{array}{ll} \text{minimise} & F(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \ldots, m, \\ & h_i(x) = 0, \quad i = 1, \ldots, p. \end{array}$$

Here $x = (x_1, \ldots, x_n)$ are decision variables. The functions $F(x), f_i$, and h_i are the objective function, inequality, and equality constraints, respectively.

Convex optimisation, a subset of optimisation, uses convex analysis theory to build methods of optimisation for difficult problems. This essay provides an overview of the theoretical foundations regarding duality, and practical methods, namely the interior point method, of convex optimisation.

2 Duality

Definition 2.1 (Convex optimisation problem)**.** A convex optimisation problem is a special case of Definition 1.1 where:

- The objective function $F(x)$ is convex.
- The inequality constraint functions $f_i(x)$ are convex.
- The equality constraint functions $h_i(x)$ are affine.

This convex optimisation problem is in the *primal* form, with inequality and equality constraints. For simplicity, we will also assume these functions are continuous and differentiable.

Duality theory is a powerful tool that underlies many algorithms in convex optimisation. By transforming a *primal* problem into a *dual* problem, we can simplify computations, gain insight into the problem, and exploit properties of the dual form.

2.1 The Lagrangian

Definition 2.2 (Lagrangian)**.** Formally, we write the **Lagrangian** of a convex optimisation problem.

$$\mathcal{L}(x, \lambda, \nu) = F(x) + \sum_{i=1}^{m} \lambda_i \, f_i(x) \; + \; \sum_{j=1}^{p} \nu_j \, h_j(x),$$

with dual variables $\lambda_i > 0$ (for inequalities) and $\nu_j \in \mathbb{R}$ (for equalities).

Intuition Behind the Lagrangian.

Suppose that we wish to minimise an objective $F(x)$ subject to constraints $f_i(x) \leq 0$ and $h_i(x) = 0$. Consider altering the objective function so that if a constraint is violated, the value of the objective increases greatly. The Lagrangian does this by adding terms of the form $\lambda_i \, f_i(x)$ and $\nu_i \, h_i(x)$ to the objective function. Where $\lambda_i \geq 0$ so the term is positive if the constraint is violated (penalising the objective) and $\nu_i \in \mathbb{R}$, since it depends on whether $h_i(x) > 0$ or $h_i(x) < 0$ holds.

Definition 2.3 (Lagrange Dual Function)**.** The *Lagrange dual function* is defined as

$$q(\lambda, \nu) \;=\; \inf_x \mathcal{L}(x, \lambda, \nu).$$

This gives us the minimum x, given two dual variables, λ and ν.

Theorem 2.4 (Weak Duality)**.** *Let p^* denote the optimal value of the primal optimisation problem. Then for any $\lambda \geq 0$ and $\nu \in \mathbb{R}^p$, we have*

$$q(\lambda, \nu) \;\leq\; p^*.$$

Proof. Suppose x^* is any *primal feasible* point. Then we have $f_i(x^*) \leq 0$ and $h_j(x^*) = 0$. For any dual feasible (λ, ν) with $\lambda_i \geq 0$, we have

$$\mathcal{L}(x^*, \lambda, \nu) \;=\; F(x^*) + \sum_{i=1}^{m} \lambda_i \, f_i(x^*) + \sum_{j=1}^{p} \nu_j \, h_j(x^*).$$

Because $f_i(x^*) \leq 0$ and $\lambda_i \geq 0$, the product $\lambda_i f_i(x^*) \leq 0$. Also, $h_j(x^*) = 0$. Hence,

$$\mathcal{L}(x^*, \lambda, \nu) \;\leq\; F(x^*).$$

Taking infimum over x in $\mathcal{L}(x, \lambda, \nu)$, we get

$$q(\lambda, \nu) \;=\; \inf_x \; \mathcal{L}(x, \lambda, \nu) \;\leq\; \mathcal{L}(x^*, \lambda, \nu) \;\leq\; F(x^*).$$

Since x^* was an arbitrary *primal feasible* point, choose x^* such that $F(x^*) = p^*$. Then

$$q(\lambda, \nu) \;\leq\; p^*.$$

□

Remark 2.5. Weak duality holds for any optimisation problem, giving us a lower bound for the primal optimum value. By finding the maximum over λ and ν, we get the best lower bound of p^*.

$$d^* \;=\; \sup_{\lambda \geq 0,\, \nu} q(\lambda, \nu) \;\leq\; p^*.$$

Where d^* is the optimum value of the dual problem.

Definition 2.6 (Dual Problem)**.** The *dual problem* is formulated as:

$$\sup_{\lambda \geq 0, \nu} q(\lambda, \nu). \tag{1}$$

Example 2.7. Consider the problem

$$\min_{x \in \mathbb{R}^2} \quad x_1^2 + x_2^2, \quad \text{subject to} \quad x_1 + x_2 = 1, \quad x_1 \geq 0.8.$$

1. Rewrite the constraints in standard form.

$$0.8 - x_1 \leq 0,$$

$$x + x_2 - 1 = 0.$$

2. Write the Lagrangian.

$$\mathcal{L}(x, \lambda, \nu) = x_1^2 + x_2^2 + \lambda(0.8 - x_1) + \nu(x_1 + x_2 - 1).$$

3. Find the Dual Function. Minimise $\mathcal{L}$ over x by setting the partial derivatives to zero:

$$\frac{\partial \mathcal{L}}{\partial x_1} = 2x_1 - \lambda + \nu = 0 \quad \Longrightarrow \quad x_1 = \frac{\lambda - \nu}{2},$$

$$\frac{\partial \mathcal{L}}{\partial x_2} = 2x_2 + \nu = 0 \quad \Longrightarrow \quad x_2 = -\frac{\nu}{2}.$$

Substitute these back into the Lagrangian:

$$\mathcal{L}(x, \lambda, \nu) = \left(\frac{\lambda - \nu}{2}\right)^2 + \left(-\frac{\nu}{2}\right)^2 + \lambda\left(0.8 - \frac{\lambda - \nu}{2}\right) + \nu\left(\frac{\lambda - \nu}{2} - \frac{\nu}{2} - 1\right).$$

After simplifying (by expanding and collecting like terms), we obtain

$$q(\lambda, \nu) = -\frac{1}{4}\lambda^2 + \frac{1}{2}\lambda\nu - \frac{1}{2}\nu^2 + 0.8\lambda - \nu.$$

4. Solve the Dual Problem.

$$d^* \;=\; \sup_{\lambda \geq 0,\, \nu \in \mathbb{R}} q(\lambda, \nu) = \sup_{\lambda \geq 0,\, \nu \in \mathbb{R}} \left(-\frac{1}{4}\lambda^2 + \frac{1}{2}\lambda\nu - \frac{1}{2}\nu^2 + 0.8\lambda - \nu\right).$$

Taking partial derivatives with respect to λ and ν and setting them to zero:

$$\frac{\partial q}{\partial \lambda} = -\frac{1}{2}\lambda + \frac{1}{2}\nu + 0.8 = 0, \quad (1)$$

$$\frac{\partial q}{\partial \nu} = \frac{1}{2}\lambda - \nu - 1 = 0. \quad (2)$$

From (1):

$$-\lambda + \nu + 1.6 = 0 \quad \Longrightarrow \quad \nu = \lambda - 1.6.$$

From (2):

$$\frac{1}{2}\lambda - (\lambda - 1.6) - 1 = 0 \quad \Longrightarrow \quad -\frac{1}{2}\lambda + 0.6 = 0,$$

so that

$$\lambda = 1.2.$$

Then,

$$\nu = 1.2 - 1.6 = -0.4.$$

Substituting $\lambda^* = 1.2$ and $\nu^* = -0.4$ into the dual function:

$$q(1.2, -0.4) = -\frac{1}{4}(1.2)^2 + \frac{1}{2}(1.2)(-0.4) - \frac{1}{2}(-0.4)^2 + 0.8(1.2) - (-0.4) = 0.68.$$

So the optimal dual value is $d^* = 0.68$.

2.2 Strong Duality

Definition 2.8 (Strong Duality)**.** *Strong duality* holds for an optimisation problem if the optimal value of the primal problem is equal to the optimal value of the dual problem, i.e.

$$p^* = d^*.$$

Remark 2.9. Unlike weak duality, this does not always hold. If there is a deficit, we call $p^* - d^*$ the *duality gap*. Additionally, if strong duality holds, we say there is *zero duality gap*.

Theorem 2.10 (Slater's Condition)**.** *Consider the* ***convex*** *optimisation problem:*

$$\min_x F(x) \quad \textit{subject to} \quad f_i(x) \le 0,\ i = 1, \ldots, m, \quad h_j(x) = 0,\ j = 1, \ldots, p,$$

Strong duality holds if there exists a point x_0 (called a strictly feasible point) such that:

$$f_i(x_0) < 0, \quad \forall i = 1, \ldots, m, \quad \textit{and} \quad h_j(x_0) = 0, \quad \forall j = 1, \ldots, p.$$

Proof. Consider the primal **convex** problem:

$$\min_x F(x) \quad \text{subject to} \quad f_i(x) \le 0,\ i = 1, \ldots, m, \quad h_j(x) = 0,\ j = 1, \ldots, p,$$

From weak duality, it is easy to see that

$$\min_x F(x) \ge \sup_{\lambda \ge 0, \nu} q(\lambda, \nu)$$

So we must show,

$$\min_{x} F(x) \leq \sup_{\lambda \geq 0, \nu} q(\lambda, \nu)$$

Consider the set:

$$A := \{(u, v, t) \in \mathbb{R}^m \times \mathbb{R}^p \times \mathbb{R} \mid \exists x, f_i(x) \leq u_i, h_j(x) = v_j, F(x) \leq t\}$$

Assuming Slater's conditions hold, the interior of A is non-empty.

Claim: A is a convex set.

This comes directly from the definition of convexity. Since $g_i(x)$ are convex, $h_j(x)$ are affine and $F(x)$ is convex, A is a convex set.

Proof. To show that A is convex, we must prove that for any two points

$$(u^1, v^1, t^1),\ (u^2, v^2, t^2) \in A,$$

and for any $\lambda \in [0, 1]$, the convex combination

$$(u^\lambda, v^\lambda, t^\lambda) := \lambda(u^1, v^1, t^1) + (1 - \lambda)(u^2, v^2, t^2)$$

also belongs to A.

Let (u^1, v^1, t^1) and (u^2, v^2, t^2) be in A with corresponding x^1 and x^2. For any $\lambda \in [0, 1]$, set

$$x^\lambda = \lambda x^1 + (1 - \lambda)x^2.$$

Then, by the convexity of f_i and F and the affinity of h_j,

$$\begin{aligned}
f_i(x^\lambda) &\leq \lambda f_i(x^1) + (1 - \lambda) f_i(x^2) \leq \lambda u_i^1 + (1 - \lambda) u_i^2, \\
h_j(x^\lambda) &= \lambda h_j(x^1) + (1 - \lambda) h_j(x^2) = \lambda v_j^1 + (1 - \lambda) v_j^2, \\
F(x^\lambda) &\leq \lambda F(x^1) + (1 - \lambda) F(x^2) \leq \lambda t^1 + (1 - \lambda) t^2.
\end{aligned}$$

Thus, the convex combination $(u^\lambda, v^\lambda, t^\lambda)$ belongs to A with corresponding x^λ, proving its convexity. □

Remark 2.11. A is the set of all possible combinations of constraint values and the objective function. Thus, by observation, primal optimal value is given by:

$$p^* = \inf\{t \mid (0, 0, t) \in A\}$$

And the dual function is:

$$\begin{aligned}
q(\lambda, \nu) &= \inf_{x} \left[F(x) + \sum_{i=1}^{m} \lambda_i f_i(x) + \sum_{j=1}^{p} \nu_j h_j(x) \right] \\
&= \inf_{x} \{\langle (\lambda, \nu, 1), (f_i(x), h_j(x), F(x)) \rangle\} \\
&= \inf_{(u,v,t)} \{\langle (\lambda, \nu, 1), (u, v, t) \rangle \mid (u, v, t) \in A\}
\end{aligned}$$

Where $\langle \cdot, \cdot \rangle$ denotes the inner product.

Additionally, consider a second set given by:

$$B := \left\{ (0,0,s) \ \in \ \mathbb{R}^m \times \mathbb{R}^p \times \mathbb{R} \ \middle| \ s < p^* \right\}.$$

Claim: A and B, do not intersect.

Proof. Consider $(u,v,t) \in A \cap B$. Since $(u,v,t) \in B$, we have $u = 0$, $v = 0$, and $t < p^*$, but since $(u,v,t) \in A$ also, $(0,0,t) \in A$, with that $t < p^*$. This contradicts our earlier remark that $p^* = \inf\{t \mid (0,0,t) \in A\}$. □

Here we will use the separating hyperplane theorem (a result from convex analysis). Essentially, it states: If two **convex** sets are disjoint, with at least one having **nonempty interior**, then there exists a hyperplane that separates them. i.e. there exists a nonzero vector

$$(\hat{\lambda}, \hat{\nu}, \hat{\mu}) \ \in \ \mathbb{R}^m \times \mathbb{R}^p \times \mathbb{R} \quad \text{and} \quad \alpha \ \in \ \mathbb{R}$$

such that

$$\langle (\hat{\lambda}, \hat{\nu}, \hat{\mu}), (u,v,t) \rangle \ \geq \ \alpha \quad \text{for all } (u,v,t) \in A, \tag{1}$$

$$\langle (\hat{\lambda}, \hat{\nu}, \hat{\mu}), (0,0,s) \rangle \ \leq \ \alpha \quad \text{for all } (0,0,s) \in B. \tag{2}$$

From (1) we can see $\hat{\lambda}_i \geq 0$ and $\hat{\mu} \geq 0$. Otherwise $\langle (\hat{\lambda}, \hat{\nu}, \hat{\mu}), (u,v,t) \rangle$ would not be bounded below.

Additionally, (2) simply states that for all $(0,0,s) \in B$ with $s < p^*$, we have

$$\langle (\hat{\lambda}, \hat{\nu}, \hat{\mu}), (0,0,s) \rangle \ = \ \hat{\mu}\, s \ \leq \ \alpha.$$

Since $s < p^*$ is arbitrary, taking the limit as $s \to p^*$ from below gives

$$\hat{\mu}\, p^* \ \leq \ \alpha.$$

Together we get,

$$\langle (\hat{\lambda}, \hat{\nu}, \hat{\mu}), (u,v,t) \rangle \ \geq \ \alpha \ \geq \ \hat{\mu}\, p^*.$$

Suppose $\hat{\mu} \neq 0$. Then dividing through by $\hat{\mu}$ gives

$$\langle (\lambda, \nu, 1), (u,v,t) \rangle \ \geq \ p^* \quad \text{for all } (u,v,t) \in A,$$

with

$$\lambda \ := \ \frac{\hat{\lambda}}{\hat{\mu}}, \qquad \nu \ := \ \frac{\hat{\nu}}{\hat{\mu}}.$$

Hence,

$$\sup_{\lambda \geq 0, \nu} q(\lambda, \nu) \ \geq \ q(\lambda, \nu) \ = \ \inf_{(u,v,t)} \{ \langle (\lambda, \nu, 1), (u,v,t) \rangle \mid (u,v,t) \in A \} \geq p^*$$

Together with weak duality,

$$d^* = \sup_{\lambda \geq 0, \nu} q(\lambda, \nu) \ = \ p^*$$

Suppose $\hat{\mu} = 0$.

Since Slater's conditions hold, the separating hyperplane theorem must hold. Intuitively, such a hyperplane must vary and depend on the objective function variable, t. If $\hat{\mu} = 0$, then the plane would be independent of t and only dependent on the constraint variables. This would make it impossible for the plane to distinguish from the set A (which contains points $(0, 0, t)$ with $t \geq p^*$) and B (which contains points $(0, 0, s)$ with $s < p^*$). The hyperplane would not be able to separate these sets, contradicting the separating hyperplane theorem, so it must be that $\hat{\mu} \neq 0$.
Rigorously, we can show how $\hat{\mu} = 0$ implies independence from t.

Suppose $\hat{\mu} = 0$. Then the separating hyperplane, using (1), reduces to

$$\langle (\hat{\lambda}, \hat{\nu}, \hat{\mu}), (u, v, t) \rangle \;=\; \hat{\lambda}^T u + \hat{\nu}^T v \;\geq\; \alpha \quad \text{for all } (u, v, t) \in A.$$

The definition of the primal optimal value,

$$p^* = \inf\{\, t \mid (0, 0, t) \in A \,\},$$

implies that for any $t \geq p^*$ the point $(0, 0, t)$ is in A, simply from the structure of A. Evaluating (1) on such a point gives

$$\hat{\lambda}^T 0 + \hat{\nu}^T 0 = 0 \geq \alpha.$$

Also, from (2) we have

$$\langle (\hat{\lambda}, \hat{\nu}, 0), (0, 0, s) \rangle = 0 \leq \alpha.$$

Thus, it follows that $\alpha = 0$ and we obtain

$$\hat{\lambda}^T u + \hat{\nu}^T v \geq 0 \quad \text{for all } (u, v, t) \in A.$$

In particular, consider points in both A and B of the form $(0, 0, z)$. For all these points, the equation of the hyperplane becomes

$$\langle (\hat{\lambda}, \hat{\nu}, 0), (0, 0, z) \rangle = 0,$$

Clearly, this plane cannot distinguish between points in A with $z \geq p^*$ and points in B with $z < p^*$. Therefore, it must be that $\hat{\mu} \neq 0$. $\square$

2.3 Karush-Kuhn-Tucker (KKT) Conditions

Theorem 2.12 (KKT Conditions)**.** *Consider the **convex** optimisation problem:*

$$\min_x F(x) \quad \textit{subject to} \quad f_i(x) \leq 0,\ i = 1, \ldots, m, \quad h_j(x) = 0,\ j = 1, \ldots, p,$$

If there is zero duality gap, then x^* and (λ^*, ν^*) are the primal and dual optimal, if and only if, they satisfy the following conditions KKT conditions*:*

1. $\nabla F(x^*) + \sum_{i=1}^{m} \lambda_i^* \nabla f_i(x^*) + \sum_{j=1}^{p} \nu_j^* \nabla h_j(x^*) = 0,$ (Stationarity)
2. $f_i(x^*) \leq 0, \quad \lambda_i^* \geq 0, \quad \forall i,$ (Primal & Dual Feasibility)
3. $h_j(x^*) = 0, \quad \forall j.$ (Equality Feasibility)
4. $\lambda_i^* f_i(x^*) = 0, \quad \forall i,$ (Complementary Slackness)

Proof. (⇒) We prove each condition in turn.

(1) Stationarity: Since x^* is an optimal (minimal) solution to the primal and (λ^*, ν^*) is optimal (supremum) for the dual with zero duality gap,

$$\mathcal{L}(x^*, \lambda, \nu) \leq \mathcal{L}(x^*, \lambda^*, \nu^*) \leq \mathcal{L}(x, \lambda^*, \nu^*) \quad \text{for all } x,\ \lambda \geq 0,\ \nu.$$

This is commonly known as the saddle point property.

In particular, x^* minimises $x \mapsto \mathcal{L}(x, \lambda^*, \nu^*)$. Thus, by linearity of the grad operator and the first-order condition,

$$\nabla_x \mathcal{L}(x^*, \lambda^*, \nu^*) = \nabla F(x^*) + \sum_{i=1}^{m} \lambda_i^* \nabla f_i(x^*) + \sum_{j=1}^{p} \nu_j^* \nabla h_j(x^*) = 0.$$

(2) & (3) Feasibility: Since x^* and λ_i^* are optimal, they must be feasible. So we have,

$$f_i(x^*) \leq 0 \quad \text{for all } i, \quad h_j(x^*) = 0 \quad \text{for all } j, \quad \lambda_i^* \geq 0, \quad \text{for all } i.$$

(4) Complementary Slackness: Recall the saddle point property tells us that x^* minimises $\mathcal{L}(x, \lambda^*, \nu^*)$.

Together with strong duality, we obtain

$$F(x^*) = q(\lambda^*, \nu^*) = \min_x \{F(x) + \lambda_i^* f(x) + \nu_i^* h(x)\} = F(x^*) + \lambda_i^* f(x^*).$$

Taking away $F(x^*)$ from both sides, we have

$$\lambda_i^* f(x^*) = 0 \quad \text{for all } i.$$

(⇐) **The converse is simple,**

Consider a triplet (x^*, λ^*, ν^*) that satisfies the KKT conditions.
By (2) and (3) they are feasible. We are left to show optimality.
Consider the Lagrangian,

$$\mathcal{L}(x^*, \lambda^*, \nu^*) = F(x^*) + \sum_i \lambda_i f_i(x^*) + \sum_j \nu_j h_j(x^*).$$

By (3) and (4),

$$\sum_i \lambda_i f_i(x^*) + \sum_j \nu_j h_j(x^*) = 0, \quad \text{so} \quad \mathcal{L}(x^*, \lambda^*, \nu^*) = F(x^*).$$

By (1),

$$\nabla_x F(x^*) = \nabla_x \mathcal{L}(x^*, \lambda^*, \nu^*) = 0.$$

By convexity of $F(x)$, x^* must minimise $F(x)$.
To prove optimality of (λ^*, ν^*), by strong duality it is now sufficient to show $q(\lambda^*, \nu^*) = F(x^*)$.

$$\begin{aligned} q(\lambda^*, \nu^*) &= \min_x \mathcal{L}(x, \lambda^*, \nu^*) && \text{(definition)} \\ &= \mathcal{L}(x^*, \lambda^*, \nu^*) && \text{(saddle point property)} \\ &= F(x^*) && \text{(as proven above)} \end{aligned}$$

□

3 Interior-Point Methods

The KKT conditions seem to be useful only when given a candidate triplet of (x, λ, ν). However, interior-point methods utilise the KKT conditions to provide a way to solve the original optimisation problem.

3.1 The Barrier Method

Directly solving the KKT conditions can be difficult. To simplify the problem, we can perturb the complementary slackness condition by replacing

$$\lambda_i\, f_i(x) = 0$$

with

$$\lambda_i\, f_i(x) = -t, \quad \text{for } t > 0.$$

Then we simply consider a sequence of solutions obtained from the perturbed problems as $t \to 0$.

First consider how this affects the other conditions. The stationarity condition of the perturbed system (for the inequality constraints) reads

$$\nabla F(x) + \sum_{i=1}^{m} \lambda_i\, \nabla f_i(x) = \nabla F(x) - \sum_{i=1}^{m} \frac{t}{f_i(x)}\, \nabla f_i(x) = 0.$$

Notice that there is a function,

$$F_t(x) \;:=\; F(x) - t \sum_{i=1}^{m} \log\bigl(-f_i(x)\bigr),$$

such that its gradient gives

$$\nabla F_t(x) = \nabla F(x) - t \sum_{i=1}^{m} \nabla\bigl[\log\bigl(-f_i(x)\bigr)\bigr] = \nabla F(x) - t \sum_{i=1}^{m} \frac{\nabla f_i(x)}{f_i(x)}.$$

Remark 3.1. In this formulation, the domain of $F_t(x)$ is restricted by $f(x) < 0$ because if $f_i(x) = 0$ for any i, the term $\log\bigl(-f_i(x)\bigr)$ becomes undefined. This restriction implicitly creates a barrier, preventing the iterates from reaching the boundary, $f_i(x) = 0$. The function $F_t(x)$ is thus commonly referred to as the *logarithmic barrier*.

Remark 3.2. It is important to note that the above reformulation by the logarithmic barrier only affects the inequality constraints $f_i(x) \leq 0$. The equality constraints $h(x) = 0$ must be enforced explicitly.

Together, the domain of $F_t(x)$ is

$$\{x \in \mathbb{R}^n \mid h(x) = 0,\; f_i(x) < 0 \text{ for all } i\},$$

and the full stationarity condition for the perturbed KKT system is

$$\nabla\Bigl(F(x) - t \sum_{i=1}^{m} \log\bigl(-f_i(x)\bigr)\Bigr) + \sum_{j=1}^{p} \nu_j\, \nabla h_j(x) = 0.$$

Thus, for a convex optimisation problem,

$$\min_x \Big\{F(x) - t\sum_{i=1}^m \log\big(-f_i(x)\big) \ : \ h(x) = 0, \ f_i(x) < 0 \text{ for all } i\Big\},$$

gives the x that minimises the Lagrangian subject to the perturbed KKT conditions, and from which the dual variables λ and ν can be recovered from $\lambda_i = -t/f_i(x)$ and the stationarity condition respectively.

Proposition 3.3. *Let*

$$x_t \ = \ \min_x \Big\{F(x) - t\sum_{i=1}^m \log\big(-f_i(x)\big) \ : \ h(x) = 0, \ f_i(x) < 0 \textit{ for all } i\Big\},$$

be the minimiser of the barrier-augmented problem, and assume that the original convex optimisation problem

$$\min_x \ F(x) \quad \textit{subject to} \quad f_i(x) \leq 0, \ h(x) = 0,$$

has zero duality gap.
Then, as $t \to 0$, the sequence $\{x_t\}_{t>0} \to x^*$, the optimal of the original convex problem. Moreover, if the dual variables corresponding to the inequality constraints are defined by

$$\lambda_{i,t} := -\frac{t}{f_i(x_t)},$$

and the dual variables ν_t associated with the equality constraints are chosen so that the full stationarity condition

$$\nabla\Big(F(x_t) - t\sum_{i=1}^m \log\big(-f_i(x_t)\big)\Big) + \sum_{j=1}^p \nu_{j,t}\,\nabla h_j(x_t) = 0$$

is satisfied, then as $t \to 0$, the triplet (x_t, λ_t, ν_t) converges to (x^*, λ^*, ν^*) that satisfies the KKT conditions for the original problem. In particular, x^* is an optimal solution of the primal problem and (λ^*, ν^*) is optimal for the dual.

Example 3.4. Consider the convex optimisation problem

$$\min_{x\in\mathbb{R}^2} \quad x_1^2 + x_2^2, \quad \text{subject to} \quad x_1 + x_2 = 1, \quad 0.8 - x_1 \leq 0, .$$

Thus

$$f(x) := 0.8 - x_1, \quad h(x) := x_1 + x_2 - 1.$$

The barrier method replaces the inequality $f(x) \leq 0$ with its logarithmic barrier,

$$F_t(x) = x_1^2 + x_2^2 - t\,\log\big(x_1 - 0.8\big),$$

Using the equality constraint, we eliminate x_2. Then the problem reduces to:

$$F_t(x_1) = x_1^2 + (1 - x_1)^2 - t\,\log(x_1 - 0.8) = 2x_1^2 - 2x_1 + 1 - t\,\log(x_1 - 0.8),$$

with $x_1 > 0.8$.
The first-order condition for optimality is given by

$$\frac{d}{dx_1}F_t(x_1) = 4x_1 - 2 - \frac{t}{x_1 - 0.8} = 0.$$

By applying a suitable minimisation method, for each $t > 0$ we obtain a unique solution $x_{1,t}$.
In the limit as $t \to 0$, the optimality condition becomes

$$(4x_1 - 2)(x_1 - 0.8) = 0.$$

Since $x_1 \geq 0.8$ is required, we must have

$$x_1^* = 0.8.$$

Then, using the equality $x_1 + x_2 = 1$, we obtain

$$x_2^* = 1 - 0.8 = 0.2.$$

Thus, the optimal primal solution is

$$x^* = (0.8, 0.2).$$

The dual variable for the inequality constraint is recovered via the perturbed complementary slackness condition:

$$\lambda_t = -\frac{t}{f(x_t)} = \frac{t}{x_{1,t} - 0.8}.$$

For small t, write $x_{1,t} = 0.8 + \delta_t$ with δ_t small. The first-order condition becomes

$$4(0.8 + \delta_t) - 2 - \frac{t}{\delta_t} = 0 \quad \Longrightarrow \quad 1.2 + 4\delta_t - \frac{t}{\delta_t} = 0.$$

Neglecting the term $4\delta_t$ for very small δ_t, we have approximately

$$\frac{t}{\delta_t} \approx 1.2,$$

so that

$$\lambda_t \approx 1.2.$$

Thus, in the limit, we set

$$\lambda^* = 1.2.$$

The stationarity condition for the full Lagrangian

$$\mathcal{L}(x, \lambda, \nu) = x_1^2 + x_2^2 + \lambda(-x_1 + 0.8) + \nu(x_1 + x_2 - 1)$$

yields, by differentiating with respect to x_1,

$$2x_1 - \lambda + \nu = 0.$$

At $x_1^* = 0.8$ and with $\lambda^* = 1.2$, we have

$$2(0.8) - 1.2 + \nu = 0 \quad \Longrightarrow \quad 1.6 - 1.2 + \nu = 0,$$

which implies

$$\nu^* = -0.4.$$

Thus, by the barrier method we obtain the optimal triplet

$$(x^*, \lambda^*, \nu^*) = \big((0.8, 0.2),\ 1.2,\ -0.4\big).$$

4 Conclusion

Duality theory and interior-point methods form a framework for tackling convex optimisation problems. When we use the Lagrangian to set up the dual problem, we view the problem from another perspective, highlighting the necessary optimality conditions and providing means to establish strong duality. The KKT conditions outline the properties an optimal solution must have, while the barrier method uses boundary and convergence theory to actually find that solution. The blend of these theoretical ideas and practical methods, are essential building blocks for modern convex optimisation.

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